

System Requirements:

This guide requires at least **12GB Vram or Higher**. I have not tested it with 8GB VRAM configurations.

Tested Hardware:

- RTX 3060 12GB - Training works but is noticeably slower
- RTX 5070 Ti 16GB - Smooth training experience with faster processing times

The performance difference between 12GB and 16GB VRAM is expected and primarily affects training speed rather than capability.

Note1: This guide does not cover LoRA training for Wan 2.2 While it's technically possible using AIToolkit, it demands high VRAM. Since you'll be using **RunPod** for Wan 2.2, it's far better to follow the video tutorial—it's simpler, clearer, and built for that setup.

- **Wan 2.2 Training Guide:** [Click Here](#)

Note2: If you wanna use Ai-toolkit in Runpod you can use my colab notebook: [ClickHere](#)

This article may be a bit lengthy, but rest assured, it is well worth your time. I will also assume that you are already familiar with the process of installing Ai-ToolKit locally.

If you don't know how to install:

- Install the prerequisites: **Git**, **Python 3.11+**, and **Node.js v22+**
- Clone the repo: `git clone https://github.com/ostris/ai-toolkit.git`
- Install Python dependencies: `pip install -r requirements.txt`
- Navigate to the UI folder and install Node dependencies: `cd ui → npm install`
- Start the UI: `npm run dev`

STEP 1: Dataset Preparation

Images: **60+ recommended, 20 minimum** — clear, sharp, varied angles/lighting

Part A — JoyCaption Setup (optimized version by me) for Image Caption

```
git clone https://github.com/official-imvoidid/Joycaption
cd Joycaption
```

Run in order: `GetConda.bat → SetEnv.bat → InstallRequirements.bat → StartTextCaptioner.bat`

1. Bulk Tab, select all images
2. Type/Select what you want the model to tag/focus on
3. Run captioning
4. Download the caption ZIP at the end
5. Unzip it → move all `.txt` files into the **same folder as your images**

Part B — Trigger Word Adder

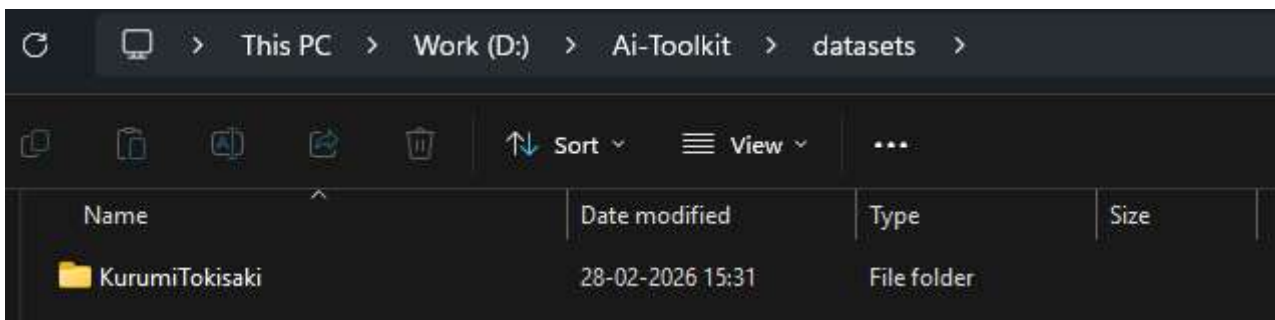
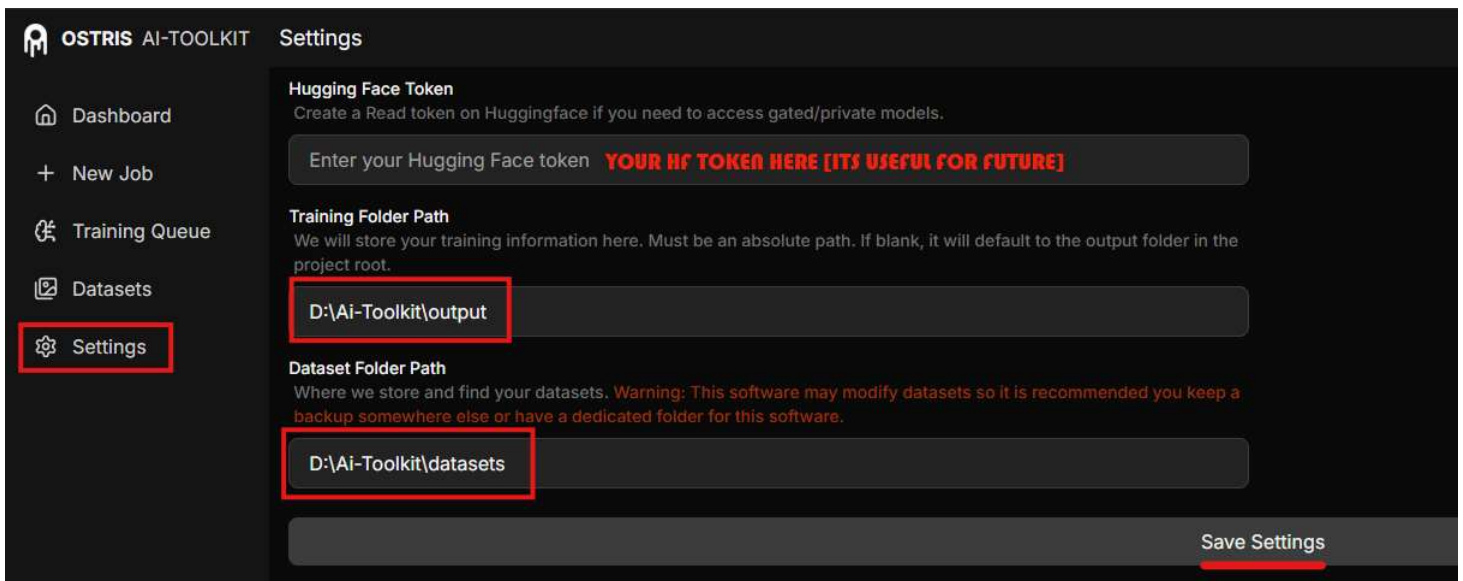
<https://github.com/official-imvoidid/Random/blob/main/TriggerWordAdder.py>

No pip install needed — pure Python. Just run `TriggerWordAdder.py` and paste the path of you folder having images and captions/tags ⚠ Critical: Type trigger word as `voidid,` (with comma+space)

-  `voidid, other-tags`
-  `voididothertags`

STEP 2: Pre-Requisite

Make Sure to move this folder containing images & caption to <Path-To-Your-Ai-Toolkit>\datasets And also to apply following settings



STEP 3: Actual Training

From Here the real training process begins, see these settings [also given as images in attachments]

OSTRIS AI-TOOLKIT < New Training Job LoRA Trainer Show Advanced Create Job

JOB Basic

Training Name: KurumiTokisaki

GPU ID: GPU #0

Trigger Word: kurusaki

MODEL

Model Architecture: Z-Image

Name or Path: Tongyi-MAI/Z-Image

Options: ☒ Low VRAM ☐ Layer Offloading Leave Default

QUANTIZATION

Transformer: float8 (default)

Text Encoder: float8 (default)

TARGET Change

Target Type: LoRA

Linear Rank: 40

SAVE

Data Type: BF16

Save Every: 500

Max Step Saves to Keep: 20

TRAINING

Batch Size: 1

Optimizer: AdamW8Bit

Learning Rate: 0.0002 Change

Weight Decay: 0.0001

Steps: 6000 Mostly Enough

Loss Type: Mean Squared Error

EMA (Exponential Moving Average): ☐ Use EMA

Text Encoder Optimizations: ☐ Unload TE ☐ Cache Text Embeddings

Regularization: ☐ Differential Output Preservation ☐ Blank Prompt Preservation

ADVANCED Does not affect training, Turn this ON

☒ Do Differential Guidance

Differential Guidance Scale: 3

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DATASETS

Dataset 1

Target Dataset: KurumiTokisaki Select Dataset

Default Caption: eg. A photo of a cat

Caption Dropout Rate: 0.05

LoRA Weight: 1

Num Repeats: 1

Settings: ☐ Cache Latents ☐ Is Regularization ☐ Flipping ☐ Flip X ☐ Flip Y

Resolutions: ☐ 256 ☒ 512 ☐ 1024 ☐ 1280 ☐ 1536

SAMPLE CHANGE

Sample Every: 500

Width: 1024

Height: 1024

Seed: 42

Advanced Sampling: ☒ Skip First Sample ☐ Force First Sample ☐ Disable Sampling

Guidance Scale: 4

Sample Steps: 25

Sample Prompts (1)

Prompt: Your-Prompt-Here

Width: 1024 (default) Height: 1024 (default) Seed: 42 (default) LoRA Scale: 1.0 (default)

Click **New Job** → **LoRA Trainer**

JOB Section

- Training Name = Your LoRA name
- Trigger Word = Same word used in your captions

MODEL Section *(leave default)*

- Architecture: Z-Image
- Model: Tongyi-MAI/Z-Image

- It will auto-download the ~30GB base model to your `.huggingface\hub` folder on first run

QUANTIZATION Section *(leave default)*

TARGET Section *(change)*

- Linear Rank: 40

SAVE Section *(change)*

- Data Type: BF16
- Save Every: 500
- Max Step Saves: 20

TRAINING Section

- Batch Size: 1 (2 if VRAM > 16GB)
- Learning Rate: 0.0002
- Steps: 6000 *(mostly enough)*

ADVANCED Section

- Turn ON **Do Differential Guidance** *(doesn't affect training quality, just enables guidance)*

DATASET Section

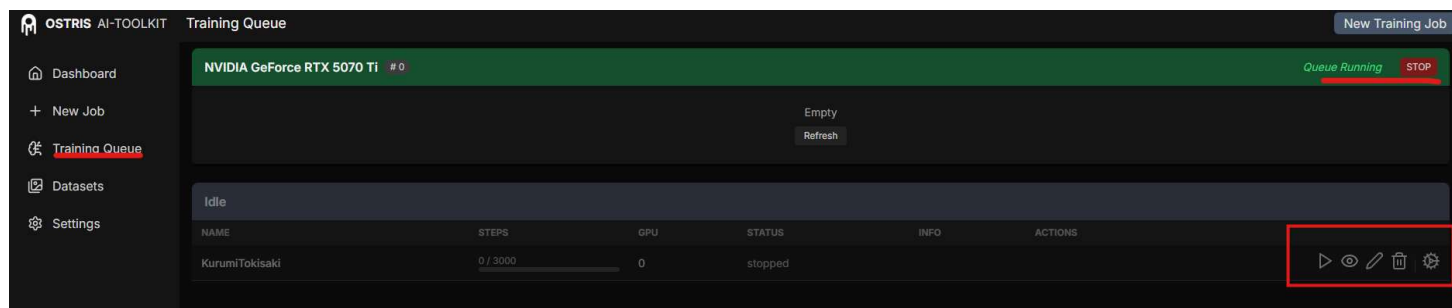
- Click **Add Dataset** → select your character's dataset
- Num Repeats: 1 (for 100+ images) / 2-3 (for ~50 images)

SAMPLE Section *(change)*

- Fill in your test prompt under **Sample Prompts** to monitor training progress visually

STEP 4: Start Training / Pause / Resume / Edit

Note: If you remove the character output folder from `\Ai-Toolkit\Output`, or delete important files, you will **not** be able to resume or edit training — it will simply restart from 0. You can, however, edit `.safetensors` files directly (e.g., `kurosaki-500.safetensors` or `kurosaki.safetensors`).



Training Locally and Enjoy Lora

-> Q/A

Q1: What tools can I use if I don't have enough VRAM?

A: If your GPU lacks sufficient VRAM, you have two excellent cloud-based alternatives. Use [RunPod](#) for LoRA and Checkpoint training, which provides powerful GPU instances on-demand. For running ComfyUI workflows in the cloud, use [GoogleColab](#) instead. You can get the Jupyter notebook for GoogleColab at my [Github](#). Both options let you train models and generate images without investing in expensive hardware.

Note: Due to recent updates on CivitAI platforms, some file attachments may no longer be visible or accessible. For all resources related to this article — along with additional tools, files, and miscellaneous resources — please refer [HERE](#)